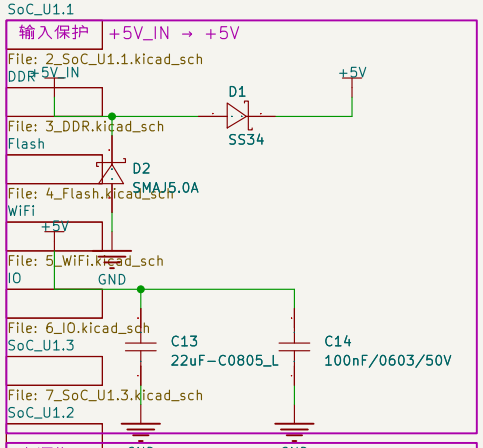
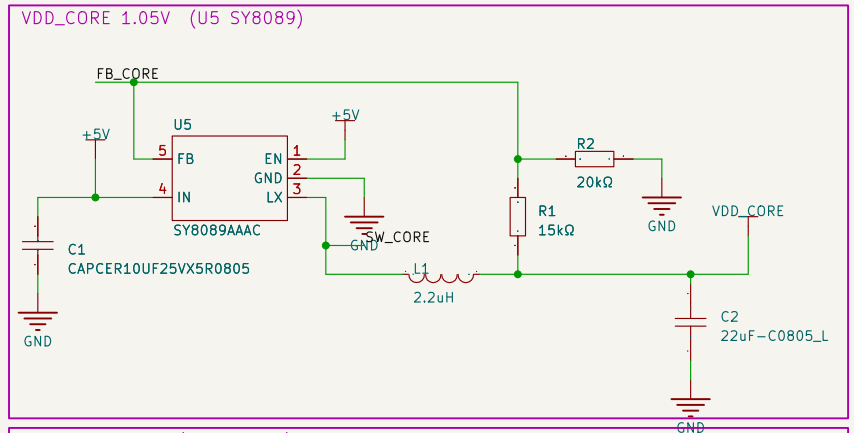
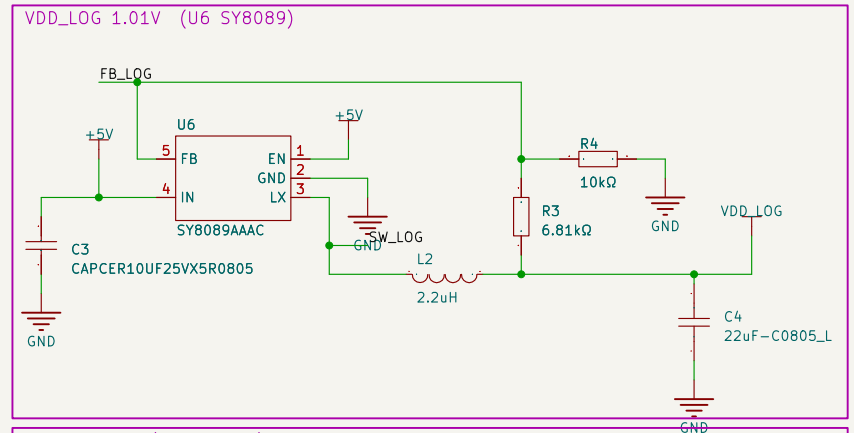
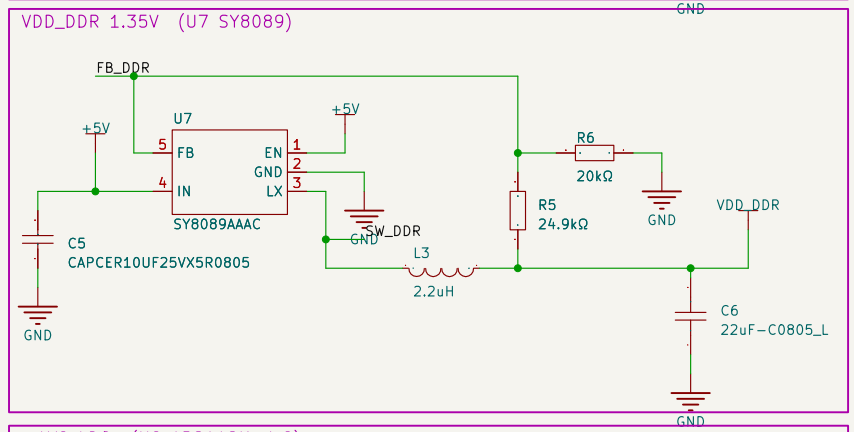
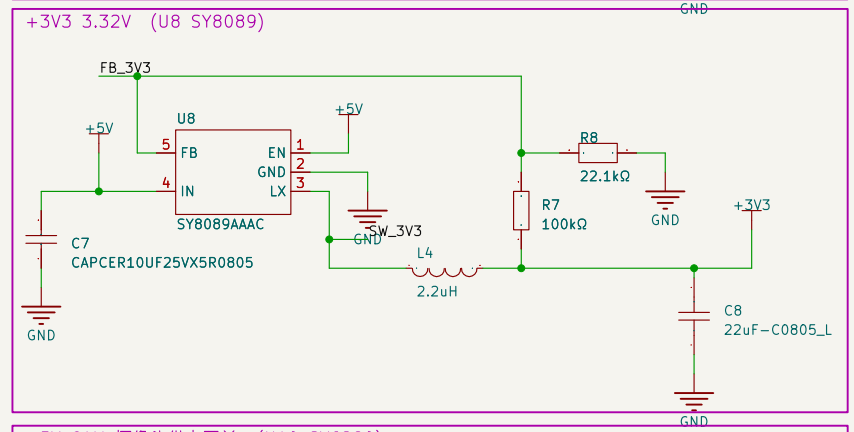
SoC_U1.1





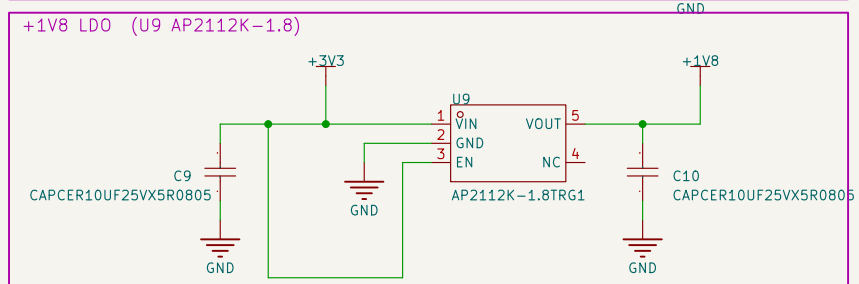


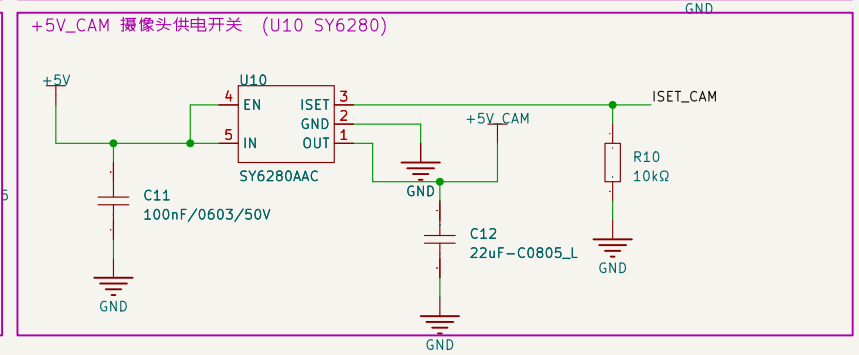




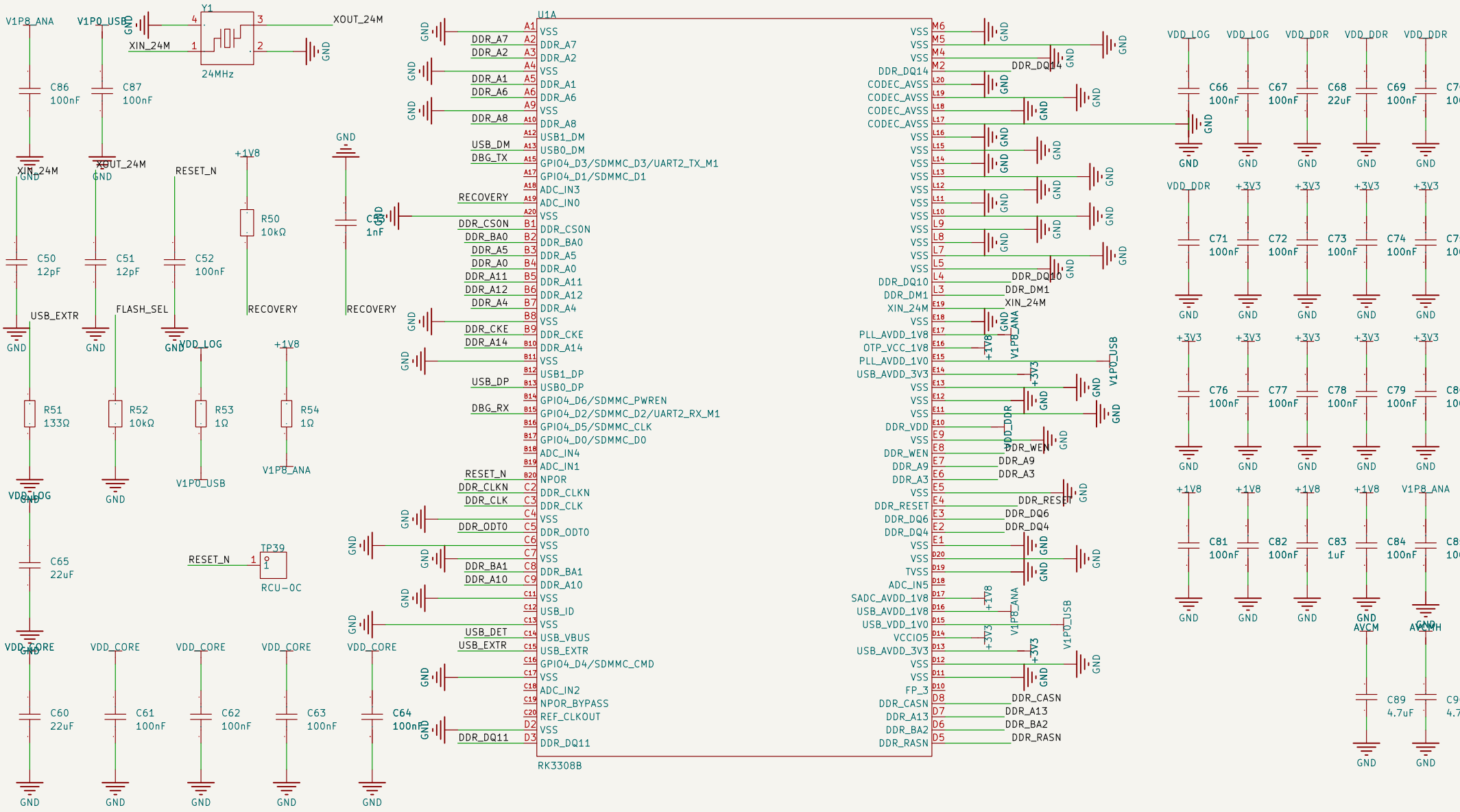
电源树

- +5V_IN → D1 SS34 → +5V (D2 SMAJ5.0A TVS)
- +5V → U5 SY8089 → VDD_CORE 1.05V
- +5V → U6 SY8089 → VDD_LOG 1.01V
- +5V → U7 SY8089 → VDD_DDR 1.35V
- +5V → U8 SY8089 → +3V3 3.32V
- +3V3 → U9 AP2112K → +1V8
- +5V → U10 SY6280 → +5V_CAM
- SY8089: VOUT = 0.6V × (1 + Rtop / Rbot)

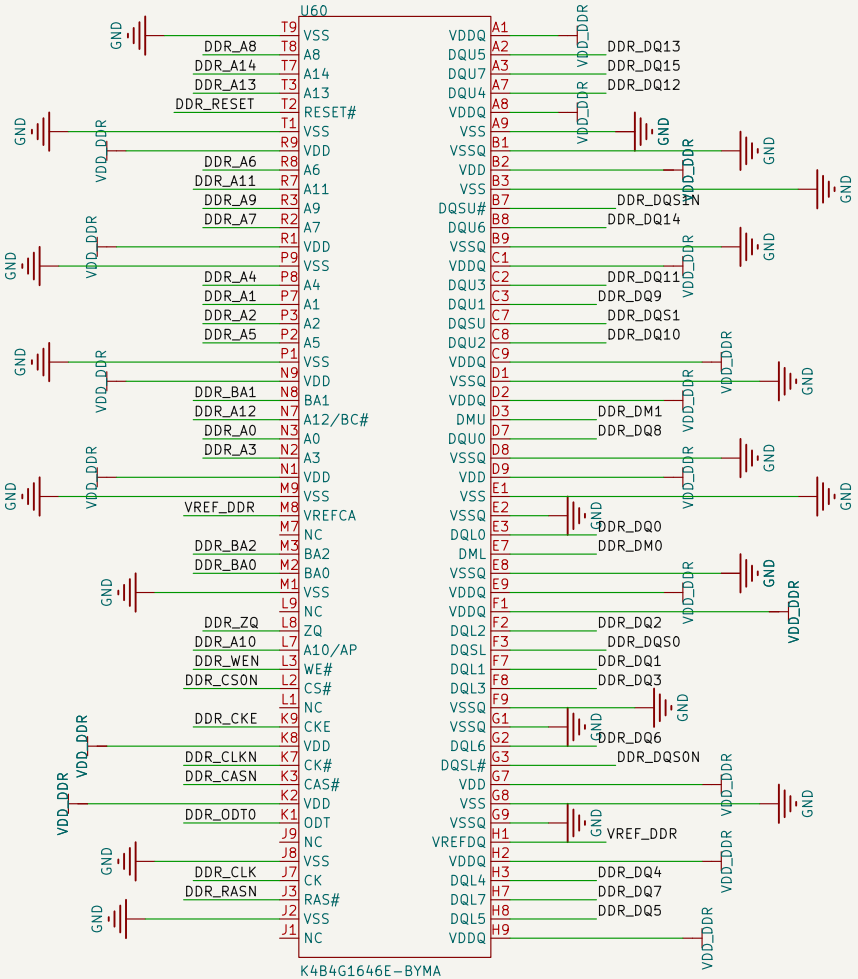


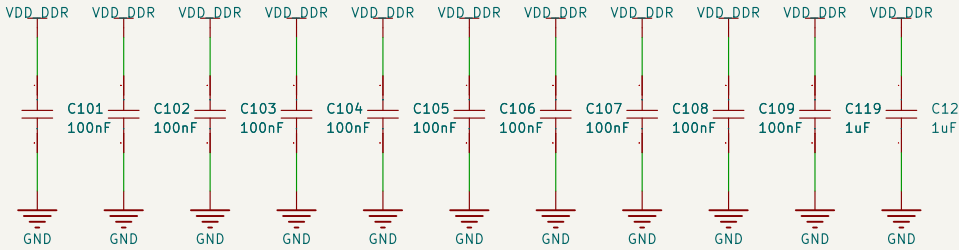


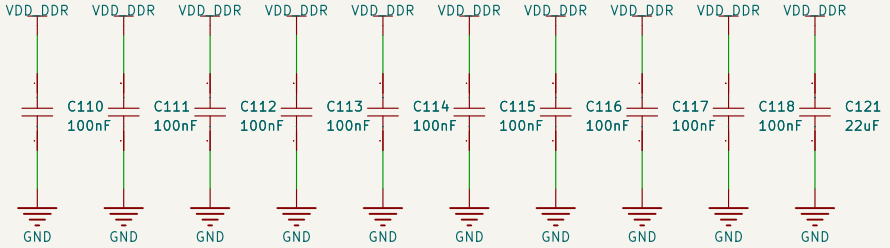
嘉立创EDA

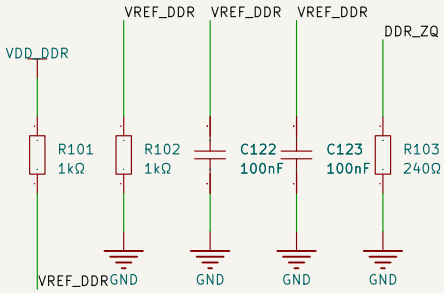

U1A
RK3308B
V1P8_ANA
V1P0_USB
XIN_24M
XOUT_24M
24MHz
Y1
RESET_N
RECOVERY
FLASH_SEL
USB_EXTR
VDD_LOG
VDD_CORE
VDD_DDR
+1V8
+3V3
GND
TP39
RCU-0C
R50 10kΩ
R51 133Ω
R52 10kΩ
R53 1Ω
R54 1Ω
C49 1nF
C50 12pF
C51 12pF
C52 100nF
C60 22uF
C61 100nF
C62 100nF
C63 100nF
C64 100nF
C65 22uF
C66 100nF
C67 100nF
C68 22uF
C69 100nF
C71 100nF
C72 100nF
C73 100nF
C74 100nF
C76 100nF
C77 100nF
C78 100nF
C79 100nF
C81 100nF
C82 100nF
C83 1uF
C84 100nF
C86 100nF
C87 100nF
C89 4.7uF
AVCM
AVCCH
DDR_A7
DDR_A2
DDR_A1
DDR_A6
DDR_A8
USB_DM
DBG_TX
RECOVERY
DDR_CSON
DDR_BA0
DDR_A5
DDR_A0
DDR_A11
DDR_A12
DDR_A4
DDR_CKE
DDR_A14
USB_DP
DBG_RX
RESET_N
DDR_CLKN
DDR_CLK
DDR_ODT0
DDR_BA1
DDR_A10
USB_ID
USB_DET
USB_EXTR
ADC_IN2
NPOR_BYPASS
REF_CLKOUT
DDR_DQ11
DDR_DQ14
DDR_DM1
XIN_24M
DDR_WEN
DDR_A9
DDR_A3
DDR_RESET
DDR_DQ6
DDR_DQ4
ADC_IN5
DDR_CASN
DDR_A13
DDR_BA2
DDR_RASN

嘉立创EDA

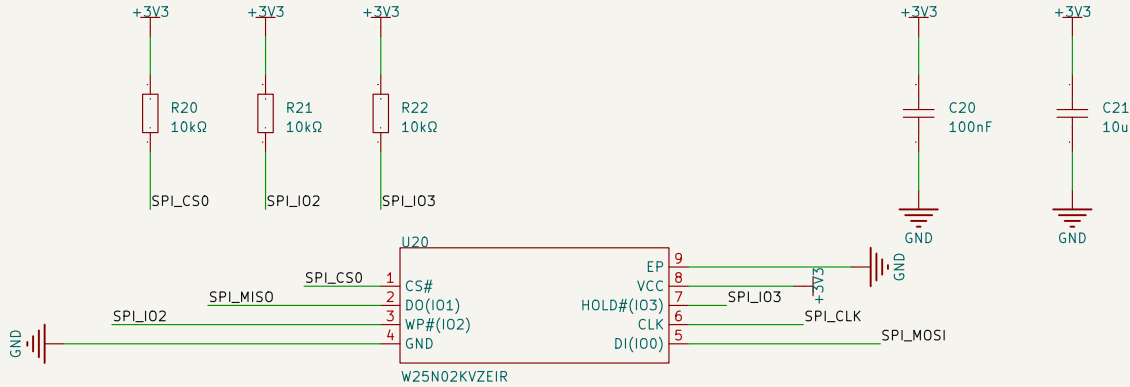

U60
K4B4G1646E-BYMA
VSS
A8
A14
A13
RESET#
VDD
A6
A11
A9
A7
A4
A1
A2
A5
BA1
A12/BC#
A0
A3
VREFCA
NC
BA2
BA0
ZQ
A10/AP
WE#
CS#
CKE
CK#
CAS#
ODT
RAS#
VDDQ
DQU5
DQU7
DQU4
DQSU#
DQU6
DQU3
DQU1
DQSU
DQU2
DMU
DQU0
DML
DQL0
DQL2
DQSL
DQL1
DQL3
DQL6
DQSL#
VREFDQ
DQL4
DQL7
DQL5
DDR_A8
DDR_A14
DDR_A13
DDR_RESET
DDR_A6
DDR_A11
DDR_A9
DDR_A7
DDR_A4
DDR_A1
DDR_A2
DDR_A5
DDR_BA1
DDR_A12
DDR_A0
DDR_A3
VREF_DDR
DDR_BA2
DDR_BA0
DDR_ZQ
DDR_A10
DDR_WEN
DDR_CS0N
DDR_CKE
DDR_CLKN
DDR_CASN
DDR_ODT0
DDR_CLK
DDR_RASN
DDR_DQ13
DDR_DQ15
DDR_DQ12
DDR_DQS1N
DDR_DQ14
DDR_DQ11
DDR_DQ9
DDR_DQS1
DDR_DQ10
DDR_DM1
DDR_DQ8
DDR_DQ0
DDR_DM0
DDR_DQ2
DDR_DQS0
DDR_DQ1
DDR_DQ3
DDR_DQ6
DDR_DQS0N
VREF_DDR
DDR_DQ4
DDR_DQ7
DDR_DQ5
VDD_DDR
GND


VDD_DDR
GND
C101 100nF
C102 100nF
C103 100nF
C104 100nF
C105 100nF
C106 100nF
C107 100nF
C108 100nF
C109 100nF
C119 1uF
C12 1uF

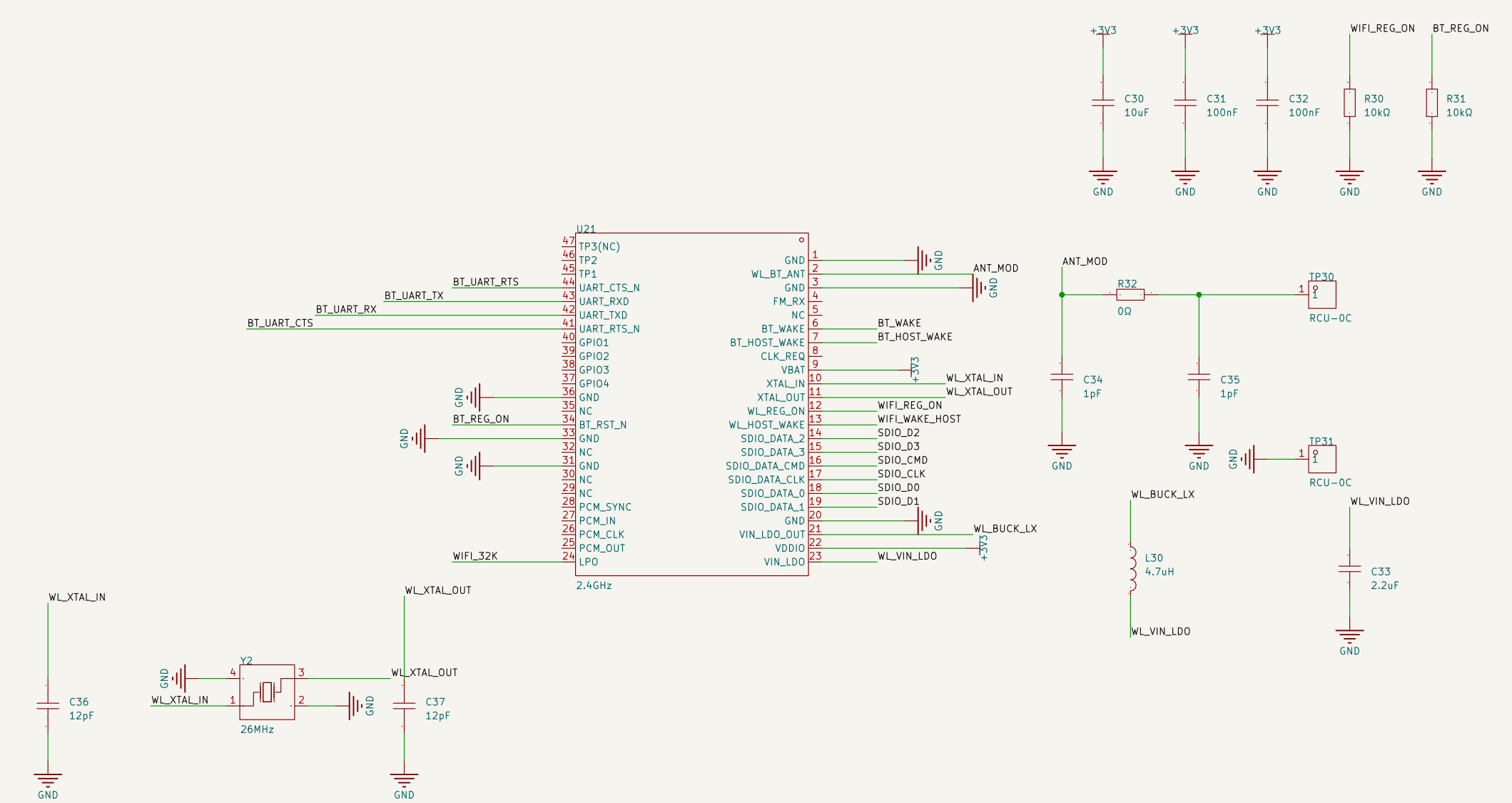

VDD_DDR
GND
C110 100nF
C111 100nF
C112 100nF
C113 100nF
C114 100nF
C115 100nF
C116 100nF
C117 100nF
C118 100nF
C121 22uF


VDD_DDR
VREF_DDR
DDR_ZQ
GND
R101 1kΩ
R102 1kΩ
C122 100nF
C123 100nF
R103 240Ω

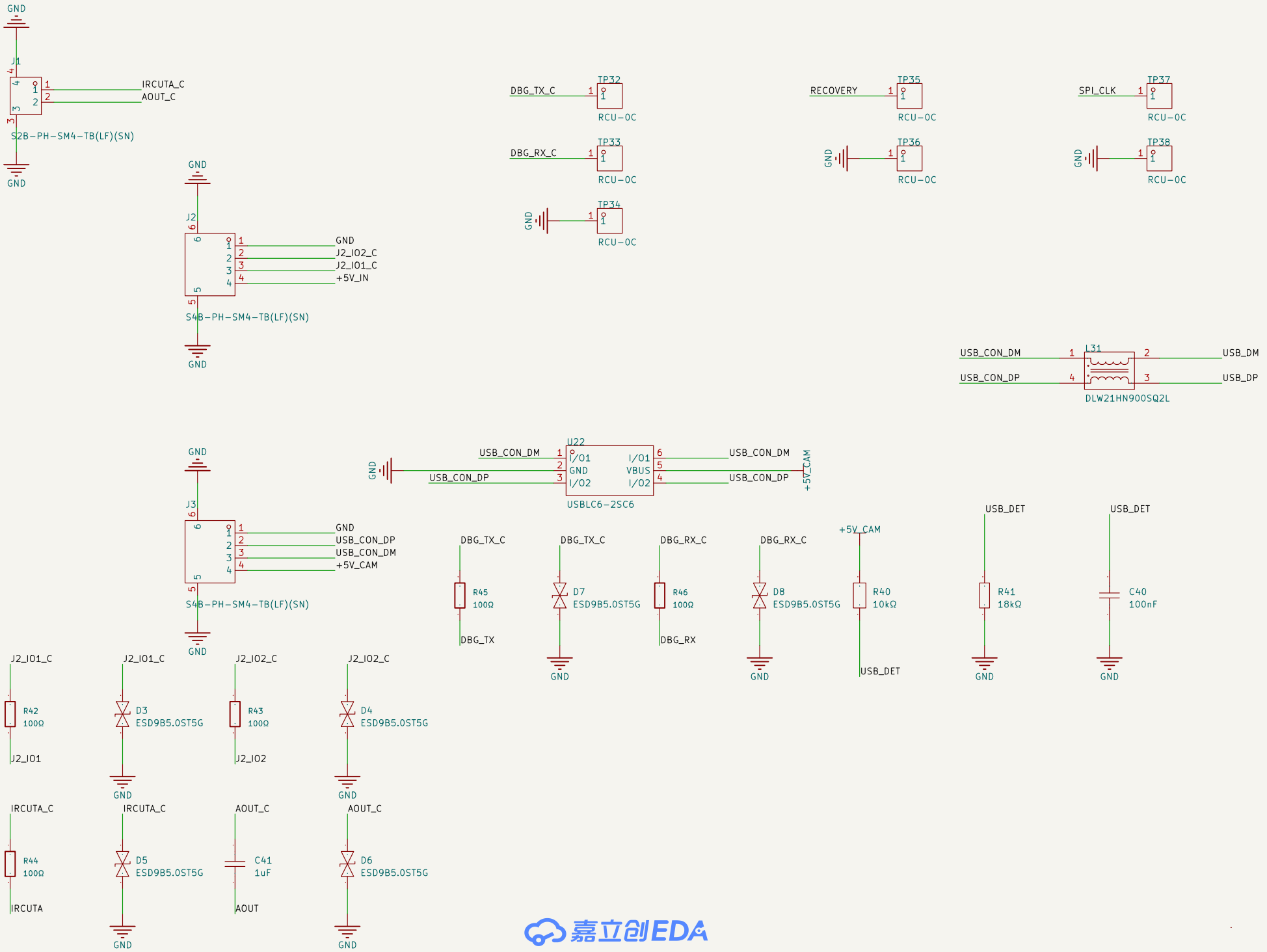
嘉立创EDA

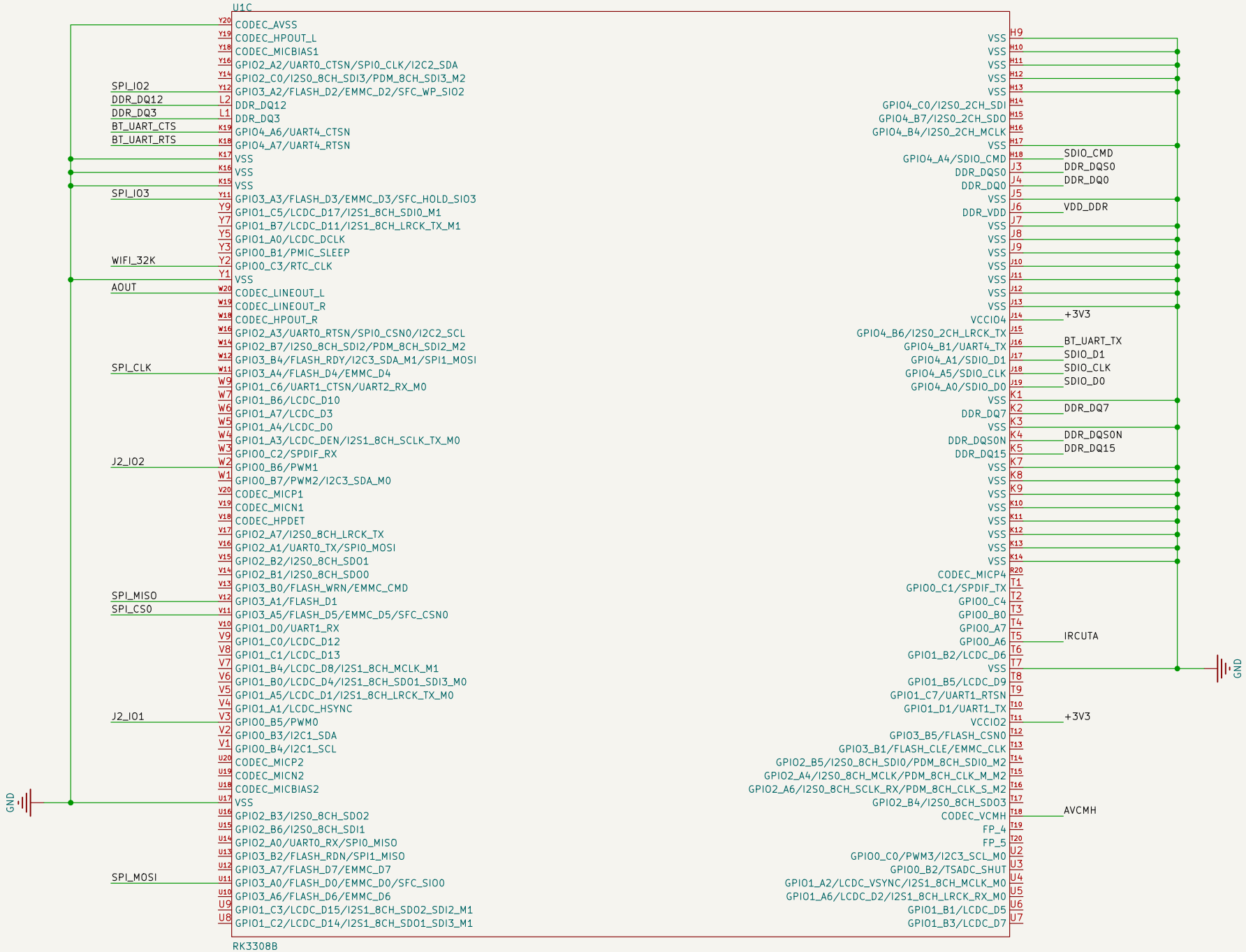

+3V3
+3V3
+3V3
R20
10kΩ
R21
10kΩ
R22
10kΩ
SPI_CS0
SPI_IO2
SPI_IO3
+3V3
+3V3
C20
100nF
C21
10u
GND
GND
U20
SPI_CS0
SPI_MISO
SPI_IO2
GND
1
2
3
4
CS#
DO(IO1)
WP#(IO2)
GND
EP
VCC
HOLD#(IO3)
CLK
DI(IO0)
9
8
7
6
5
+3V3
GND
SPI_IO3
SPI_CLK
SPI_MOSI
W25N02KVZEIR

嘉立创EDA

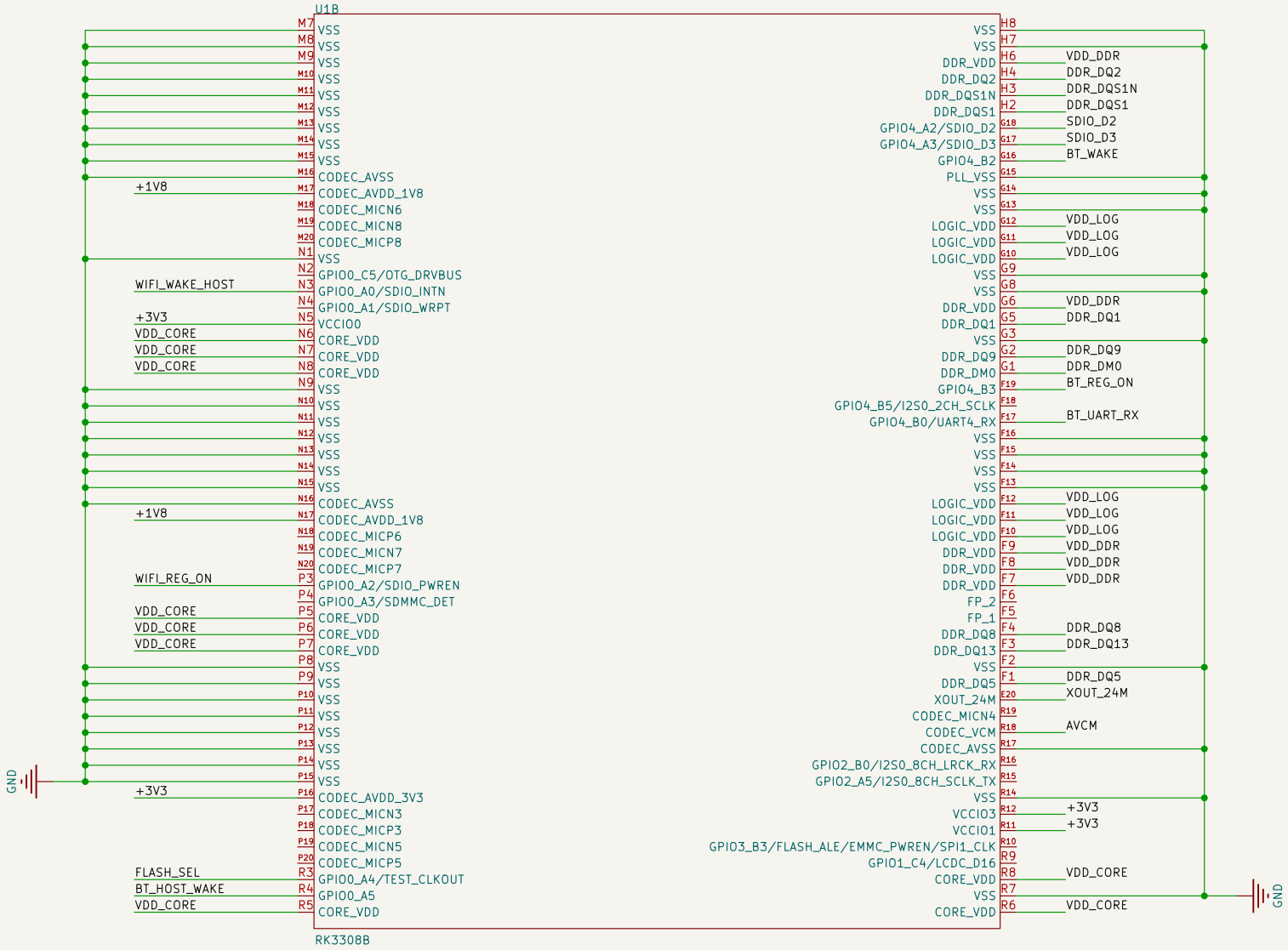

+3V3
C30
10uF
GND
+3V3
C31
100nF
GND
+3V3
C32
100nF
GND
WIFI_REG_ON
R30
10kΩ
GND
BT_REG_ON
R31
10kΩ
GND
U21
TP3(NC)
TP2
TP1
UART_CTS_N
UART_RXD
UART_TXD
UART_RTS_N
GPIO1
GPIO2
GPIO3
GPIO4
GND
NC
BT_RST_N
GND
NC
GND
NC
NC
PCM_SYNC
PCM_IN
PCM_CLK
PCM_OUT
LPO
2.4GHz
GND
WL_BT_ANT
GND
FM_RX
NC
BT_WAKE
BT_HOST_WAKE
CLK_REQ
VBAT
XTAL_IN
XTAL_OUT
WL_REG_ON
WL_HOST_WAKE
SDIO_DATA_2
SDIO_DATA_3
SDIO_DATA_CMD
SDIO_DATA_CLK
SDIO_DATA_0
SDIO_DATA_1
GND
VIN_LDO_OUT
VDDIO
VIN_LDO
BT_UART_RTS
BT_UART_TX
BT_UART_RX
BT_UART_CTS
GND
BT_REG_ON
GND
GND
WIFI_32K
GND
ANT_MOD
GND
BT_WAKE
BT_HOST_WAKE
+3V3
WL_XTAL_IN
WL_XTAL_OUT
WIFI_REG_ON
WIFI_WAKE_HOST
SDIO_D2
SDIO_D3
SDIO_CMD
SDIO_CLK
SDIO_D0
SDIO_D1
GND
WL_BUCK_LX
+3V3
WL_VIN_LDO
ANT_MOD
R32
0Ω
TP30
RCU-0C
C34
1pF
GND
C35
1pF
GND
GND
TP31
RCU-0C
WL_BUCK_LX
L30
4.7uH
WL_VIN_LDO
WL_VIN_LDO
C33
2.2uF
GND
WL_XTAL_IN
C36
12pF
GND
WL_XTAL_OUT
GND
Y2
WL_XTAL_IN
WL_XTAL_OUT
26MHz
GND
C37
12pF
GND

嘉立创EDA


GND
J1
IRCUTA_C
AOUT_C
S2B-PH-SM4-TB(LF)(SN)
GND
DBG_TX_C
TP32
RCU-0C
DBG_RX_C
TP33
RCU-0C
GND
TP34
RCU-0C
RECOVERY
TP35
RCU-0C
GND
TP36
RCU-0C
SPI_CLK
TP37
RCU-0C
GND
TP38
RCU-0C
GND
J2
GND
J2_IO2_C
J2_IO1_C
+5V_IN
S4B-PH-SM4-TB(LF)(SN)
GND
USB_CON_DM
L31
USB_DM
USB_CON_DP
USB_DP
DLW21HN900SQ2L
GND
J3
GND
USB_CON_DP
USB_CON_DM
+5V_CAM
S4B-PH-SM4-TB(LF)(SN)
GND
U22
USB_CON_DM
I/O1
GND
I/O2
I/O1
VBUS
I/O2
USB_CON_DM
USB_CON_DP
+5V_CAM
USBLC6-2SC6
DBG_TX_C
R45
100Ω
DBG_TX
DBG_TX_C
D7
ESD9B5.0ST5G
GND
DBG_RX_C
R46
100Ω
DBG_RX
DBG_RX_C
D8
ESD9B5.0ST5G
GND
+5V_CAM
R40
10kΩ
USB_DET
USB_DET
R41
18kΩ
GND
USB_DET
C40
100nF
GND
J2_IO1_C
R42
100Ω
J2_IO1
J2_IO1_C
D3
ESD9B5.0ST5G
GND
J2_IO2_C
R43
100Ω
J2_IO2
J2_IO2_C
D4
ESD9B5.0ST5G
GND
IRCUTA_C
R44
100Ω
IRCUTA
IRCUTA_C
D5
ESD9B5.0ST5G
GND
AOUT_C
C41
1uF
AOUT
AOUT_C
D6
ESD9B5.0ST5G
GND
嘉立创EDA


U1C
CODEC_AVSS
CODEC_HPOUT_L
CODEC_MICBIAS1
GPIO2_A2/UART0_CTSN/SPI0_CLK/I2C2_SDA
GPIO2_C0/I2S0_8CH_SDI3/PDM_8CH_SDI3_M2
GPIO3_A2/FLASH_D2/EMMC_D2/SFC_WP_SIO2
DDR_DQ12
DDR_DQ3
GPIO4_A6/UART4_CTSN
GPIO4_A7/UART4_RTSN
VSS
VSS
VSS
GPIO3_A3/FLASH_D3/EMMC_D3/SFC_HOLD_SIO3
GPIO1_C5/LCDC_D17/I2S1_8CH_SDI0_M1
GPIO1_B7/LCDC_D11/I2S1_8CH_LRCK_TX_M1
GPIO1_A0/LCDC_DCLK
GPIO0_B1/PMIC_SLEEP
GPIO0_C3/RTC_CLK
VSS
CODEC_LINEOUT_L
CODEC_LINEOUT_R
CODEC_HPOUT_R
GPIO2_A3/UART0_RTSN/SPI0_CSN0/I2C2_SCL
GPIO2_B7/I2S0_8CH_SDI2/PDM_8CH_SDI2_M2
GPIO3_B4/FLASH_RDY/I2C3_SDA_M1/SPI1_MOSI
GPIO3_A4/FLASH_D4/EMMC_D4
GPIO1_C6/UART1_CTSN/UART2_RX_M0
GPIO1_B6/LCDC_D10
GPIO1_A7/LCDC_D3
GPIO1_A4/LCDC_D0
GPIO1_A3/LCDC_DEN/I2S1_8CH_SCLK_TX_M0
GPIO0_C2/SPDIF_RX
GPIO0_B6/PWM1
GPIO0_B7/PWM2/I2C3_SDA_M0
CODEC_MICP1
CODEC_MICN1
CODEC_HPDET
GPIO2_A7/I2S0_8CH_LRCK_TX
GPIO2_A1/UART0_TX/SPI0_MOSI
GPIO2_B2/I2S0_8CH_SDO1
GPIO2_B1/I2S0_8CH_SDO0
GPIO3_B0/FLASH_WRN/EMMC_CMD
GPIO3_A1/FLASH_D1
GPIO3_A5/FLASH_D5/EMMC_D5/SFC_CSN0
GPIO1_D0/UART1_RX
GPIO1_C0/LCDC_D12
GPIO1_C1/LCDC_D13
GPIO1_B4/LCDC_D8/I2S1_8CH_MCLK_M1
GPIO1_B0/LCDC_D4/I2S1_8CH_SDO1_SDI3_M0
GPIO1_A5/LCDC_D1/I2S1_8CH_LRCK_TX_M0
GPIO1_A1/LCDC_HSYNC
GPIO0_B5/PWM0
GPIO0_B3/I2C1_SDA
GPIO0_B4/I2C1_SCL
CODEC_MICP2
CODEC_MICN2
CODEC_MICBIAS2
VSS
GPIO2_B3/I2S0_8CH_SDO2
GPIO2_B6/I2S0_8CH_SDI1
GPIO2_A0/UART0_RX/SPI0_MISO
GPIO3_B2/FLASH_RDN/SPI1_MISO
GPIO3_A7/FLASH_D7/EMMC_D7
GPIO3_A0/FLASH_D0/EMMC_D0/SFC_SIO0
GPIO3_A6/FLASH_D6/EMMC_D6
GPIO1_C3/LCDC_D15/I2S1_8CH_SDO2_SDI2_M1
GPIO1_C2/LCDC_D14/I2S1_8CH_SDO1_SDI3_M1
RK3308B
VSS
VSS
VSS
VSS
VSS
GPIO4_C0/I2S0_2CH_SDI
GPIO4_B7/I2S0_2CH_SDO
GPIO4_B4/I2S0_2CH_MCLK
VSS
GPIO4_A4/SDIO_CMD
DDR_DQS0
DDR_DQ0
VSS
DDR_VDD
VSS
VSS
VSS
VSS
VSS
VSS
VSS
VCCIO4
GPIO4_B6/I2S0_2CH_LRCK_TX
GPIO4_B1/UART4_TX
GPIO4_A1/SDIO_D1
GPIO4_A5/SDIO_CLK
GPIO4_A0/SDIO_D0
VSS
DDR_DQ7
VSS
DDR_DQS0N
DDR_DQ15
VSS
VSS
VSS
VSS
VSS
VSS
VSS
VSS
CODEC_MICP4
GPIO0_C1/SPDIF_TX
GPIO0_C4
GPIO0_B0
GPIO0_A7
GPIO0_A6
GPIO1_B2/LCDC_D6
VSS
GPIO1_B5/LCDC_D9
GPIO1_C7/UART1_RTSN
GPIO1_D1/UART1_TX
VCCIO2
GPIO3_B5/FLASH_CSN0
GPIO3_B1/FLASH_CLE/EMMC_CLK
GPIO2_B5/I2S0_8CH_SDI0/PDM_8CH_SDI0_M2
GPIO2_A4/I2S0_8CH_MCLK/PDM_8CH_CLK_M_M2
GPIO2_A6/I2S0_8CH_SCLK_RX/PDM_8CH_CLK_S_M2
GPIO2_B4/I2S0_8CH_SDO3
CODEC_VCMH
FP_4
FP_5
GPIO0_C0/PWM3/I2C3_SCL_M0
GPIO0_B2/TSADC_SHUT
GPIO1_A2/LCDC_VSYNC/I2S1_8CH_MCLK_M0
GPIO1_A6/LCDC_D2/I2S1_8CH_LRCK_RX_M0
GPIO1_B1/LCDC_D5
GPIO1_B3/LCDC_D7
SPI_IO2
DDR_DQ12
DDR_DQ3
BT_UART_CTS
BT_UART_RTS
SPI_IO3
WIFI_32K
AOUT
SPI_CLK
J2_IO2
SPI_MISO
SPI_CS0
J2_IO1
SPI_MOSI
GND
SDIO_CMD
DDR_DQS0
DDR_DQ0
VDD_DDR
+3V3
BT_UART_TX
SDIO_D1
SDIO_CLK
SDIO_D0
DDR_DQ7
DDR_DQS0N
DDR_DQ15
IRCUTA
+3V3
AVCMH
GND

嘉立创EDA


U1B
M7 VSS
M8 VSS
M9 VSS
M10 VSS
M11 VSS
M12 VSS
M13 VSS
M14 VSS
M15 VSS
M16 CODEC_AVSS
+1V8 M17 CODEC_AVDD_1V8
M18 CODEC_MICN6
M19 CODEC_MICN8
M20 CODEC_MICP8
N1 VSS
N2 GPIO0_C5/OTG_DRVBUS
WIFI_WAKE_HOST N3 GPIO0_A0/SDIO_INTN
N4 GPIO0_A1/SDIO_WRPT
+3V3 N5 VCCIO0
VDD_CORE N6 CORE_VDD
VDD_CORE N7 CORE_VDD
VDD_CORE N8 CORE_VDD
N9 VSS
N10 VSS
N11 VSS
N12 VSS
N13 VSS
N14 VSS
N15 VSS
N16 CODEC_AVSS
+1V8 N17 CODEC_AVDD_1V8
N18 CODEC_MICP6
N19 CODEC_MICN7
N20 CODEC_MICP7
WIFI_REG_ON P3 GPIO0_A2/SDIO_PWREN
P4 GPIO0_A3/SDMMC_DET
VDD_CORE P5 CORE_VDD
VDD_CORE P6 CORE_VDD
VDD_CORE P7 CORE_VDD
P8 VSS
P9 VSS
P10 VSS
P11 VSS
P12 VSS
P13 VSS
P14 VSS
P15 VSS
+3V3 P16 CODEC_AVDD_3V3
P17 CODEC_MICN3
P18 CODEC_MICP3
P19 CODEC_MICN5
P20 CODEC_MICP5
FLASH_SEL R3 GPIO0_A4/TEST_CLKOUT
BT_HOST_WAKE R4 GPIO0_A5
VDD_CORE R5 CORE_VDD
H8 VSS
H7 VSS
H6 VDD_DDR
H4 DDR_DQ2
H3 DDR_DQS1N
H2 DDR_DQS1
G18 SDIO_D2
G17 SDIO_D3
G16 BT_WAKE
G15 PLL_VSS
G14 VSS
G13 VSS
G12 VDD_LOG
G11 VDD_LOG
G10 VDD_LOG
G9 VSS
G8 VSS
G6 VDD_DDR
G5 DDR_DQ1
G3 VSS
G2 DDR_DQ9
G1 DDR_DM0
F19 BT_REG_ON
F18
F17 BT_UART_RX
F16 VSS
F15 VSS
F14 VSS
F13 VSS
F12 VDD_LOG
F11 VDD_LOG
F10 VDD_LOG
F9 VDD_DDR
F8 VDD_DDR
F7 VDD_DDR
F6
F5
F4 DDR_DQ8
F3 DDR_DQ13
F2 VSS
F1 DDR_DQ5
E20 XOUT_24M
R19
R18 AVCM
R17
R16
R15
R14 VSS
R12 +3V3
R11 +3V3
R10
R9
R8 VDD_CORE
R7 VSS
R6 VDD_CORE
DDR_VDD
DDR_DQ2
DDR_DQS1N
DDR_DQS1
GPIO4_A2/SDIO_D2
GPIO4_A3/SDIO_D3
GPIO4_B2
PLL_VSS
VSS
VSS
LOGIC_VDD
LOGIC_VDD
LOGIC_VDD
VSS
VSS
DDR_VDD
DDR_DQ1
VSS
DDR_DQ9
DDR_DM0
GPIO4_B3
GPIO4_B5/I2S0_2CH_SCLK
GPIO4_B0/UART4_RX
VSS
VSS
VSS
VSS
LOGIC_VDD
LOGIC_VDD
LOGIC_VDD
DDR_VDD
DDR_VDD
DDR_VDD
FP_2
FP_1
DDR_DQ8
DDR_DQ13
VSS
DDR_DQ5
XOUT_24M
CODEC_MICN4
CODEC_VCM
CODEC_AVSS
GPIO2_B0/I2S0_8CH_LRCK_RX
GPIO2_A5/I2S0_8CH_SCLK_TX
VSS
VCCIO3
VCCIO1
GPIO3_B3/FLASH_ALE/EMMC_PWREN/SPI1_CLK
GPIO1_C4/LCDC_D16
CORE_VDD
VSS
CORE_VDD
RK3308B
GND
GND

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